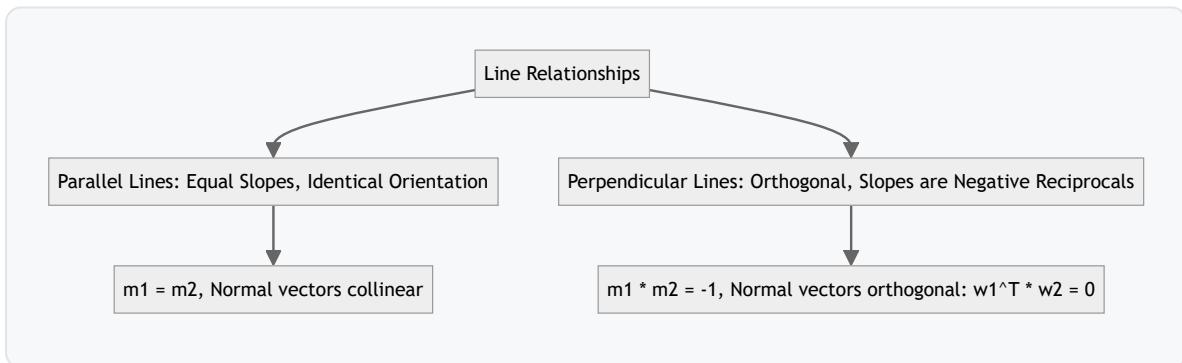


Day 21 - Lecture 21.4: Parallel & Perpendicular Lines

[Python 3.10+](#)[Jupyter Notebook](#)[PDF notes.pdf](#)[Language: English](#) | [हिंदी / Hinglish](#) | [Back to Lecture 21 Module](#)

1. Overview & Geometric Orientation

Understanding the conditions for **parallelism** and **orthogonality (perpendicularity)** is essential for coordinate geometry, gradient ascent/descent, normal vector calculations, and Support Vector Machine margin construction.



2. Mathematical Formalism

1. Parallel Lines:

Two non-vertical lines $L_1 : y = m_1x + c_1$ and $L_2 : y = m_2x + c_2$ are parallel if and only if their slopes are strictly equal:

$$m_1 = m_2 \quad (c_1 \neq c_2)$$

In general form $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

2. Perpendicular (Orthogonal) Lines:

Two non-vertical lines L_1 and L_2 are mutually perpendicular if and only if the product of their slopes equals -1 :

$$m_1 \cdot m_2 = -1 \iff m_2 = -\frac{1}{m_1}$$

In general form:

$$\left(-\frac{a_1}{b_1}\right) \cdot \left(-\frac{a_2}{b_2}\right) = -1 \implies a_1 a_2 + b_1 b_2 = 0$$

Notice that this is the **vector dot product** of their normal vectors $\mathbf{w}_1 = [a_1, b_1]^T$ and $\mathbf{w}_2 = [a_2, b_2]^T$:

$$\mathbf{w}_1^T \mathbf{w}_2 = 0 \iff L_1 \perp L_2$$

3. Python Implementation

```
import numpy as np

# Line 1: 3x - 4y + 5 = 0 -> m1 = 3/4 = 0.75
a1, b1, c1 = 3.0, -4.0, 5.0
m1 = -a1 / b1

# Parallel Line 2: 3x - 4y - 10 = 0 -> m2 = 3/4 = 0.75
a2, b2, c2 = 3.0, -4.0, -10.0
m2 = -a2 / b2

# Perpendicular Line 3: 4x + 3y + 2 = 0 -> m3 = -4/3 = -1.333
a3, b3, c3 = 4.0, 3.0, 2.0
m3 = -a3 / b3

print(f"Slope m1: {m1:.4f}, Slope m2: {m2:.4f}, Slope m3: {m3:.4f}")
print(f"L1 and L2 Parallel: {np.isclose(m1, m2)}")
print(f"L1 and L3 Perpendicular: {np.isclose(m1 * m3, -1.0)}")

# Vector dot product orthogonality check
w1 = np.array([a1, b1])
w3 = np.array([a3, b3])
print(f"Dot product w1 . w3: {np.dot(w1, w3):.1f} (Must be 0)")
```

4. Key Takeaways & Interview Points

- **Dot Product Rule:** Two lines (or hyperplanes) are perpendicular if and only if the dot product of their normal weight vectors equals zero ($\mathbf{w}_1^T \mathbf{w}_2 = 0$).
- **Support Vector Machines:** The decision boundary and the margin planes are strictly parallel hyperplanes separated by bias offsets.